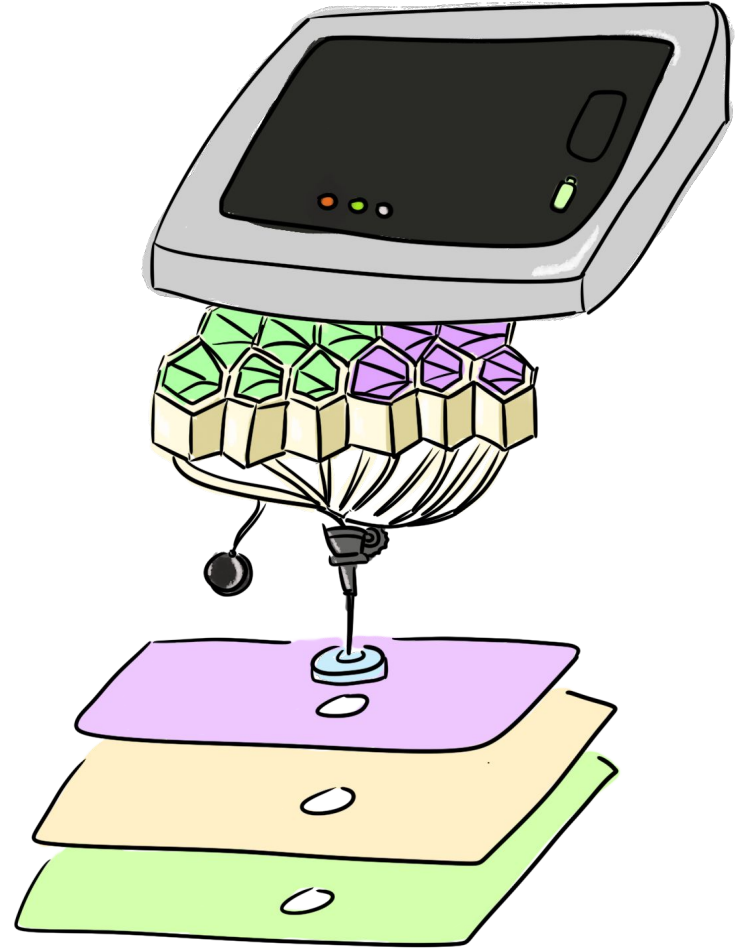


# ThinkDose

Origami-Inspired Smart  
Patch



# User Needs and Limitations Assessment

---

01



## Clinical Limitations:

Incompatible drugs cannot be co-delivered (e.g., hydrophobic and hydrophilic pairs: Doxorubicin + Paclitaxel for cancer, Adalimumab + Methotrexate in arthritis). Most conventional lack sufficient flexibility for precise, multi-drug administration.

---

02

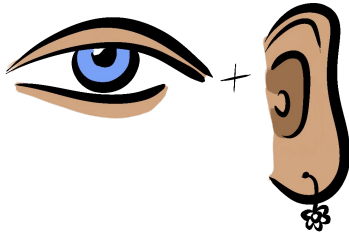


## Technical Limitations:

Large reservoirs cause pressure changes during emptying, leading to dosing inaccuracies and inconsistent flow. This makes it difficult for conventional devices to deliver precise, repeatable doses, especially at sub-microlitre levels.

---

03



## User Experience Limitations:

Current devices are bulky and uncomfortable for patients. Furthermore, they have complex interfaces that require patient training and assistance.

---

# Value Proposition



**Safe co-delivery** of incompatible drugs  
(hydrophilic + hydrophobic).



**Combination therapy** in chronic disease  
(e.g., API + pain management)



**Slim, body-contoured wearable** patient  
comfort + discretion

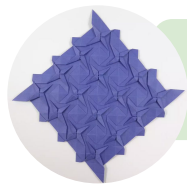


**Precise & repeatable dosing:** from  
microdoses to large volumes.

# The Design Process: From a Flat Multi-Fold Patch to 3D Honeycomb Structure

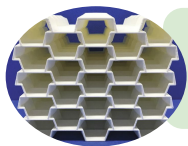
We were looking for a shape that would enable us to segment incompatible drugs and deliver anything from the microdose of 0.1 mL to the full 25 mL

## Preliminary Concepts



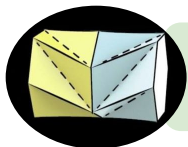
**Idea 1:** Squares and Stars Tessellations

Too compact to store 25mL



**Idea 2:** Compressible Tachi-Miura Polyhedron

Couldn't selectively actuate drugs



**Idea 3:** Kresling Origami

Could collapse catastrophically without structural support

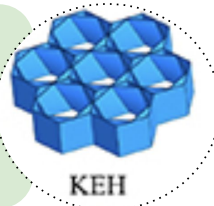
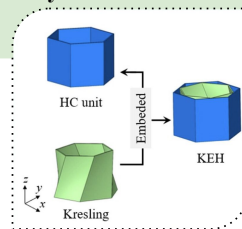
Exploration and Refinement Phase

Integration and Idea Consolidation Phase



**Intermediate Concept:**  
Multiple Kresling Dipoles

**Final Idea:**  
Kresling-Embedded Honeycomb



The rationale is discussed on slide 9.



# The Novelty of the Device



## Origami-Inspired Honeycomb Reservoir

Unique microfluidic architecture segments drugs, enabling precision dosing and scalable volume delivery. It can co-administer incompatible drug classes (e.g., hydrophobic + hydrophilic pairs, and API + pain management drugs pairs) with high precision.



## Co-delivery of Incompatible Drug

Opens a new therapeutic market by safely combining drug classes (e.g., oncology, immunotherapy) previously unable to be stored or delivered together.



## Flexible and Body-Contouring

Wearable design conforms to patients, improving comfort and adherence for long-term therapies in both hospital and home settings.



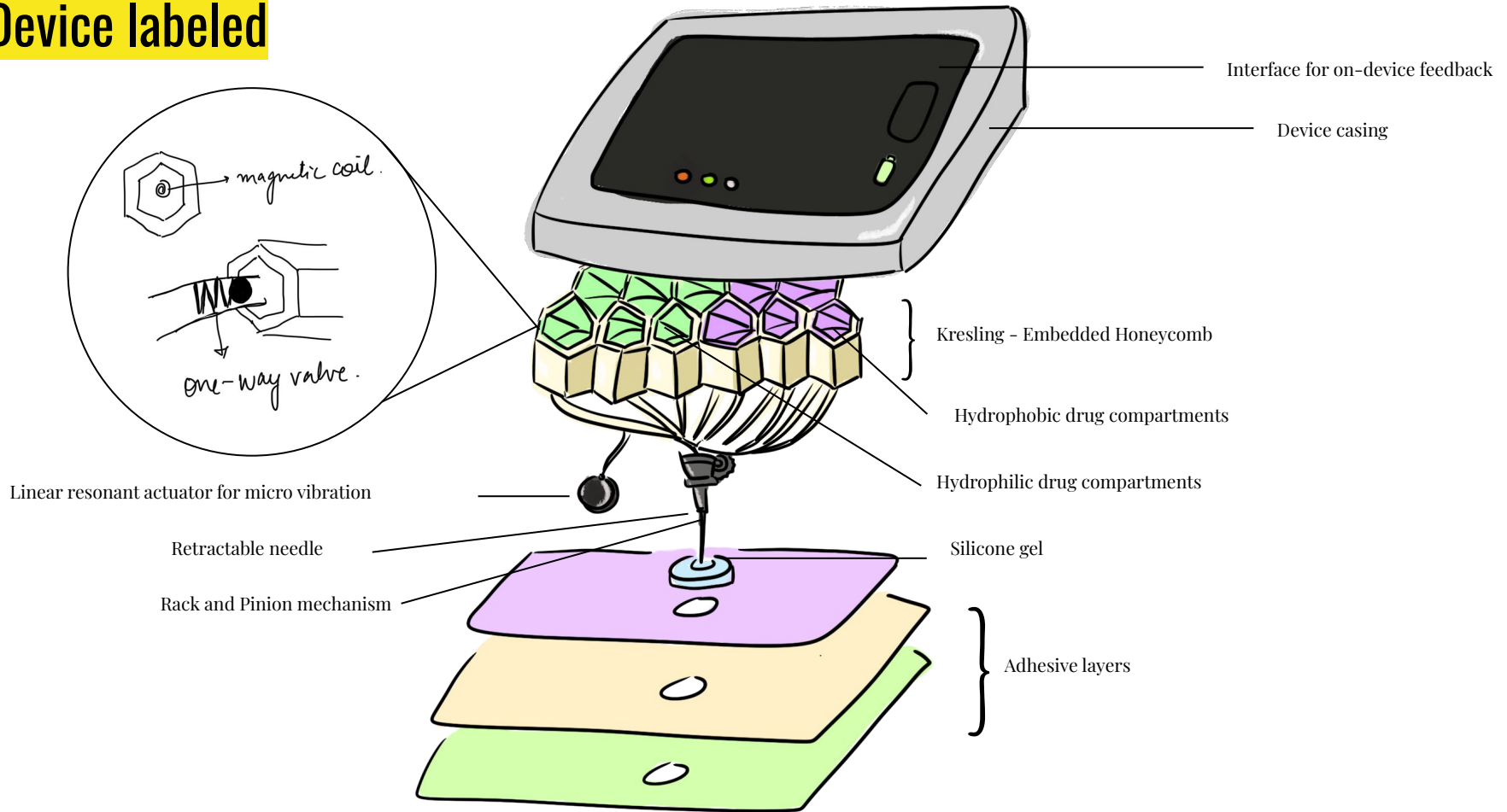
## Reusable IoT Device with Replaceable Drug Reservoir

Smart, connected platform enables remote monitoring, programmable dosing, and sustainability through modular, refillable reservoirs, driving down cost per treatment.

# The Overview of the Device

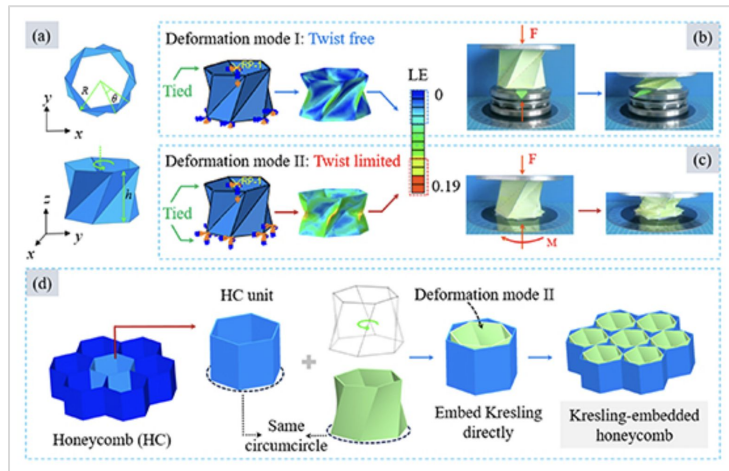


# Device labeled

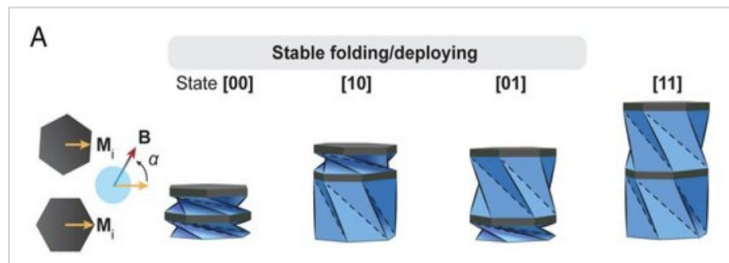


# The Technology

# Scalable, Multi-Drug Reservoir for Next-Gen Targeted Therapies



Yang, H., Liu, Y., Lu, H., Tao, R., Wei, X. and Fu, Y.-Q., 2024. 3D-printable Kresling-embedded honeycomb metamaterials with optimized energy absorption capability. *Smart Materials and Structures*, 33(12), p. 125008. <https://doi.org/10.1088/1361-665X/ad8c04>



Huang, L., Xie, T. and Yin, L., (2024). Magnetically-actuated intracorporeal biopsy robot based on Kresling Origami. *Theoretical and Applied Mechanics Letters*, 15(1), p. 100500. DOI: 10.1016/j.taml.2024.100500.

Reimagining the drug reservoir of an OBDS through Kresling-embedded honeycomb metamaterials. Inspired by *3D-printable Kresling-embedded honeycomb metamaterials with optimised energy absorption capability* (Yang et al., 2024) and *Magnetically-actuated intracorporeal biopsy robot based on Kresling Origami* (Huang et al., 2024).

## The Key Features:

### Multi-Compartment Design

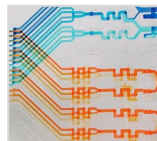
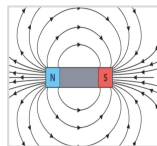
27-cell honeycomb architecture enabling drug co-delivery of incompatible drugs in a single dose. 26 cells dedicated to the API and 1 for the cannula.

### Precise Magnetic Actuation

An energised coil creates a magnetic torque on a kresling-embedded microplate, triggering a snap-through instability in the kresling structure to generate a localised pressure pulse

### Drug Expulsion

The pressure pulse expels the drug payload into a microfluidic channel. The dose rate and volume are modulated by current amplitude and field frequency.



# Rationale for this Approach

SCIENCE ADVANCES | RESEARCH ARTICLE

## ENGINEERING

### Soft robotic origami crawler

Qiji Ze<sup>1</sup>†, Shuai Wu<sup>1</sup>†, Jun Nishikawa<sup>2</sup>, Jize Dai<sup>2</sup>, Yue Sun<sup>2,3</sup>, Sophie Leanza<sup>2</sup>, Cole Zemelka<sup>2</sup>, Larissa S. Novelino<sup>4</sup>, Glaucio H. Paulino<sup>4,5,6\*</sup>, Ruike Renee Zhao<sup>1,2\*</sup>

Biomimetic soft robotic crawlers have attracted extensive attention in various engineering fields, owing to their adaptivity to different terrains. Earthworm-like crawlers realize locomotion through in-plane contraction, while inchworm-like crawlers exhibit out-of-plane bending-based motions. Although in-plane contraction crawlers demonstrate effective motion in confined spaces, miniaturization is challenging because of limited actuation methods and complex structures. Here, we report a magnetically actuated small-scale origami crawler with in-plane contraction. The contraction mechanism is achieved through a four-unit Kresling origami assembly consisting of two Kresling dipoles with two-level symmetry. Magnetic actuation is used to provide appropriate torque distribution, enabling a small-scale and untethered robot with both crawling and steering capabilities. The crawler can overcome large resistances from severely confined spaces by its anisotropic and magnetically tunable structural stiffness. The multifunctionality of the crawler is explored by using the internal cavity of the crawler for drug storage and release. The magnetic origami crawler can potentially serve as a minimally invasive device for biomedical applications.

## INTRODUCTION

Crawling motion is a navigation strategy that is commonly observed in animals, especially in worms. In these animals, body contraction paired with inhomogeneous friction between the surfaces

an alternative is crawlers made of stimuli-responsive soft materials that generate contraction via large deformation of the soft crawler body (25, 26), which usually has a relatively simple structure that permits small-scale design. Some recent crawlers with a millimeter-

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Ze, Q., Wu, S., Nishikawa, J., Dai, J., Sun, Y., Leanza, S., Zemelka, C., Novelino, L.S., Paulino, G.H. & Zhao, R.R., 2022. Soft robotic origami crawler. *Science Advances*, 8(13), eabm7834. doi:10.1126/sciadv.abm7834.

## The Structural Rationale:

The honeycomb scaffold protects the otherwise fragile Kresling origami units from local buckling or damage, making the system robust against skin deformation, movement, and mechanical stresses encountered in ambulatory patients.



Ze, Q., Dong, L., Ge, Y., Wang, Y., Zhang, W., Wu, S., Guo, Z., Deng, Z. and Wu, X., (2022). Spinning-enabled wireless amphibious origami millirobot. *Nature Communications*, 13(1), p.3118. DOI: 10.1038/s41467-022-30802-w

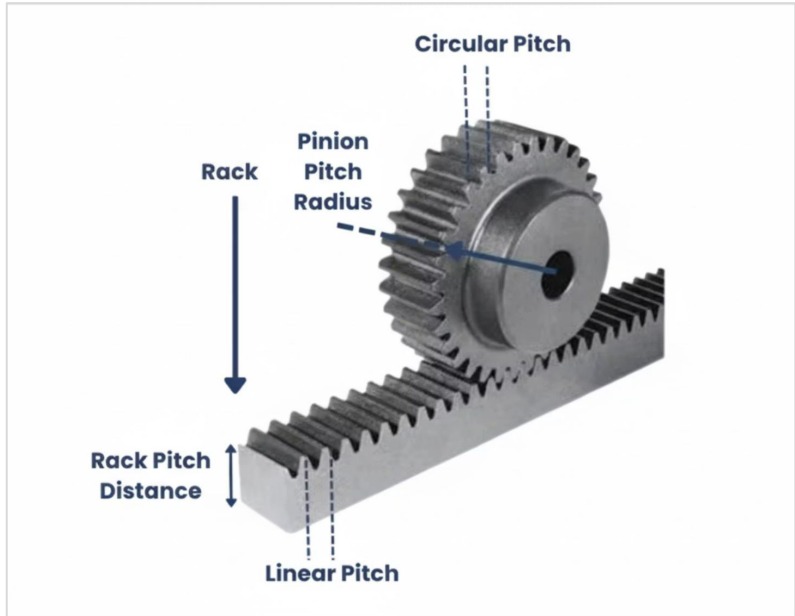
Magnetically actuated Kresling structures have been demonstrated in untethered milli-robots, where external rotating magnetic fields induce twisting, spinning, or peristaltic motion. This design adapts the principle to a stationary, wearable platform, with the LSR Kresling lattice serving as both actuator and drug reservoir, and the Ni<sub>80</sub>Fe<sub>20</sub> micro-plate converting magnetic energy to mechanical motion in a biocompatible system.

## How precise dosing is achieved:

Planar micro-coils in the hub generate rotating magnetic fields to actuate selected KEH reservoirs, which collapse under torque to expel drug through micro-valves. Dose volume is set by coil current amplitude and rate by actuation frequency; [Delivery rate= volume per actuation, set by amplitude x frequency of actuations] Each unit operates independently without affecting neighbouring reservoirs, and real-time management logs dose, drug type, and cell ID with millilitre precision for clinician-programmed personalised therapy.

# Needle Insertion and Retraction: Rack-and-Pinion Mechanism

- 1 Ultra-fine 27G soft cannula (10mm, depth-limited)
- 2 Spring-fired introducer with micro rack-and-pinion control
- 3 Dual-interlock safety system
- 4 Rapid auto-retraction and lock (< 200ms)
- 5 Intuitive LED guidance



# Integrated Sensor Features



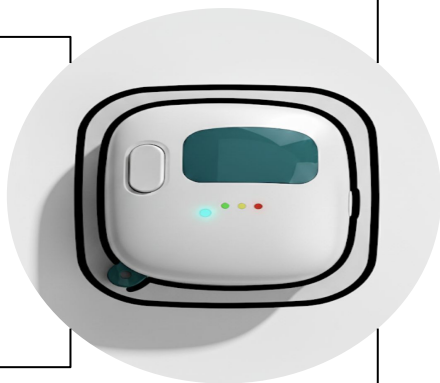
## Physiological Sensors

PPG sensor + pressure pad + GSR (sweat) that ensures dosing only occurs when skin secure skin contact and correct pressure are confirmed.



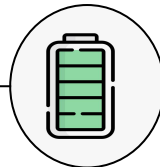
## Motion and Safety Monitoring

6-axis accelerometer + gyroscope that detects motion patterns and activity. Skin temperature for additional physiological context.



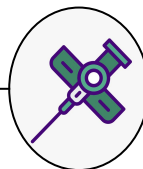
## Device Performance and Delivery Verification

Coil current and back EMF sensing + piezo admittance measurement which infers dose delivery and cannula occlusion



## Power and Connectivity Telemetry

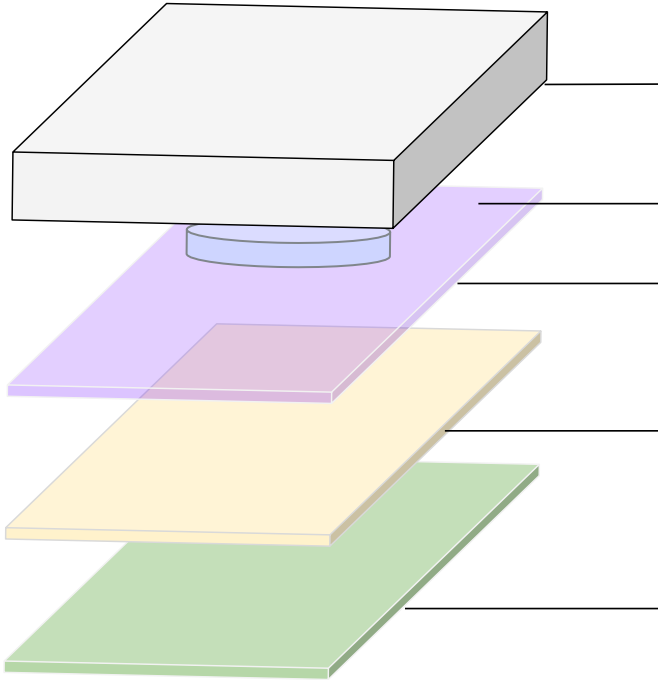
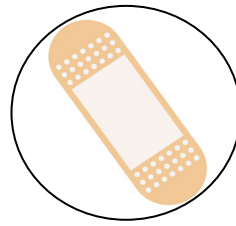
Battery and charging monitoring, offering real-time status and alerts for low charge.  
Dock-case open detection (Hall sensor), offering device usage tracking and security.  
GPS via phone integration, for remote monitoring and adherence insight.



## Cannula Control and Feedback

Insertion/ retraction switches + encoders ensure precise mechanical tracking of cannula status

# The Device Adhesive Layers



## **The Device**

Thermoplastic Polyurethane which conforms to body

## **Silicone Gel Island** (*shear reduction layer*)

## **Backing/ Substrate Pad** (*structural support layer*)

Sits above both adhesive zones and holds them together, flexible and biocompatible polymer film with high tear resistance and moisture vapour moisture transmission.

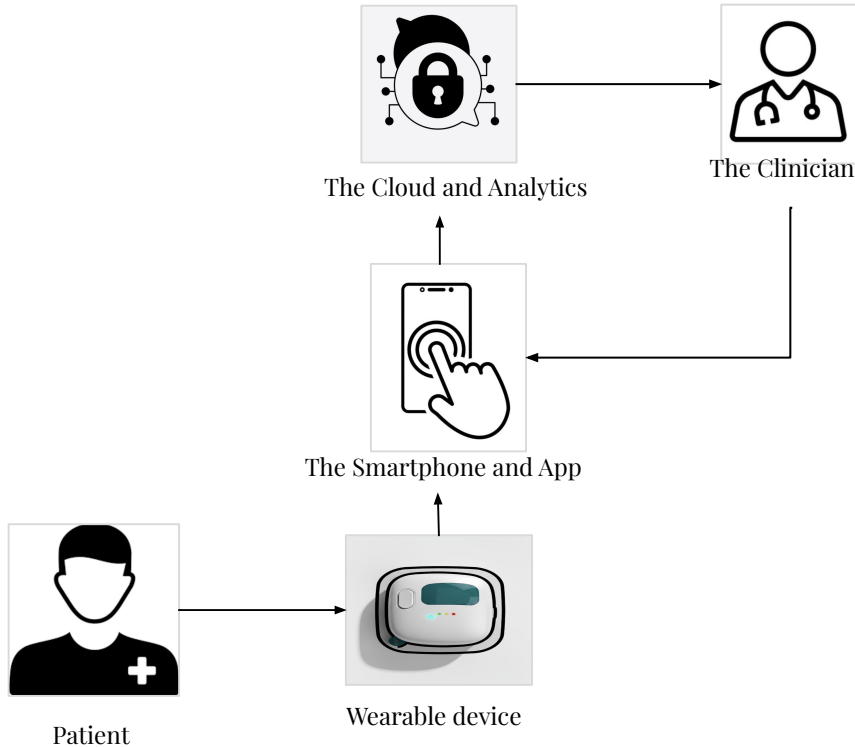
## **Outer Reinforcement Zone** (*skin interface*)

Stronger adhesive “ring” around the inner silicone zone, distributes shear and peel forces, ensuring device in sweat, movement, and extended wear.

## **Inner Silicone Zone** (*skin interface*)

Medical grade silicone adhesive, made from acrylic, stable yet low-peel force bond, consistent tack for multiple days without residue.

# The Smart Device and Feedback System



## The Key Features:



### Real-time Dose Management

Real-time dose management by tracking dose per cell, drug class, and cell ID with mL precision



### Flexible Scheduling and Control

Supports custom schedules, overrides, and caregiver/ admin modes for multi-user or dependent patients



### Intelligent alerts and safety notifications

Detects occlusion, off-skin application, low battery, and expired cartridge



### Compliance and Analytics Dashboard

Measure adherence (%), wear time, site changes, and wear time. Actionable insight for clinicians, researchers, and care teams.



### Secure Data Transmission and Storage

End-to-End encrypted sync: BLE - smartphone - cloud. Compliant with HIPAA, ISO 27001, and GDPR standards.

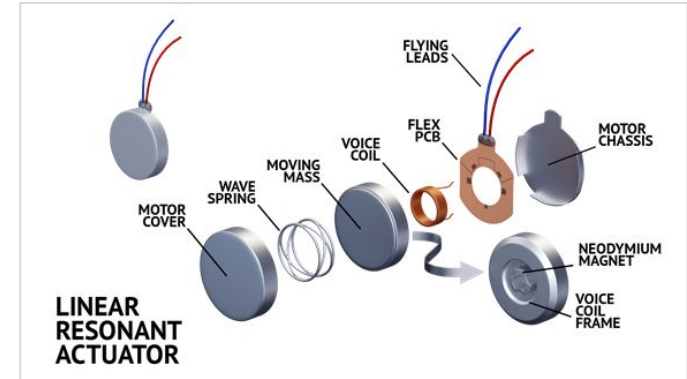
# App Accessibility

## Smart User Interface & Feedback Systems

- **Voice-Enabled Control (App):** Integrated voice recognition within the companion app enables hands-free operation, scheduling, and override commands, improving accessibility for elderly and mobility-limited users.
- **Multimodal Feedback (Device):** On-device sound and light indicators signal initiation and completion of drug delivery, while alarm cues (auditory + visual) provide immediate notification of faults such as occlusion, off-skin detection, or cartridge expiry.
- **Closed-Loop Manufacturer Feedback:** Automatic error reporting transmits anonymised malfunction data directly from the app to the manufacturer's secure cloud, enabling predictive maintenance, rapid root-cause analysis, and continuous device improvement.

# Optional Pain Relief Components for Patient Comfort

- **Micro - vibration during insertion:** Linear resonant actuator at  $\sim 100$  Hz with 0.5 g amplitude runs for  $\sim 10$  seconds timed to needle deployment activating large fiber sensory pathways to reduce perceived pain
  - Shahidi Bonjar, A.H. (2011). Syringe micro vibrator: A new device to reduce pain and anxiety of intraoral injections. Annals of Surgical Innovation and Research, 5(1), 1. <https://doi.org/10.1186/1750-1164-5-1>
- **Silicone gel cushion:** Gel island under inner silicone zone for shear relief



# Materials & Manufacturing

# Materials

## Device Casing

- Material: Thermoplastic Polyurethane
- Rationale:
  - Flexible
  - Biocompatibility: Meets ISO 10993 and USP Class VI standards for irritability

## The Flexible Kresling Unit (Drug Reservoir)

- Material: Medical-Grade Liquid Silicone Rubber (LSR) –USP Class VI.
- Rationale:
  - Biocompatibility: Extremely inert and safe for direct contact with the API.
  - Durability and Flexibility: Engineering for millions of compression cycles without material fatigue, ensuring reliable and repeatable dosing.
  - Precision: Ideal for manufacturing complex geometries like the Kresling pattern.

## Rigid Honeycomb scaffold (Protective Frame)

- Material: Medical-Grade Polycarbonate (PC).
- Rationale:
  - Structural Integrity: Provides the necessary strength and rigidity to protect the drug units and ensure actuation force is directed efficiently.
  - Stability: A well-understood, robust, and commonly used material in FDA-approved medical devices.

# Manufacturing

## Step 1: Component Fabrication

- **Kresling Reservoir Units**
  - Fabricated using Liquid Injection Moulding (LIM) of medical-grade liquid silicone rubber (LSR), enabling highly detailed, repeatable, and biocompatible drug-contact parts. LIM ensures precision wall thickness, elasticity, and robustness for millions of cycles without fatigue.
- **Honeycomb Scaffold**
  - Early prototypes produced with high-precision CNC machining to validate tolerances and assembly. For scaling, the scaffold transitions to injection-moulded medical-grade polycarbonate (PC), reducing per-unit cost and enabling high-volume, ISO-compliant manufacture.
- **Casing**
  - SLS 3D printing with Thermoplastic Polyurethane

## Step 2: Actuation Mechanism

- **Magnetic Actuation**
  - Permalloy thin foil patterned into disc using photolithography and wet-etching
  - Microplates are attached to Kresling apex using medical-grade epoxy
- **Planar Microcoils**
  - Planar micro-coils are fabricated on flexible polymer substrates (polyimide) using photolithographic PCB processes, conductive paths formed in copper deposited by inkjet printing.

## Step 3: Sterile, Assembly, Filling, and Packaging

- Conducted in ISO 7/8 cleanroom under GMP standards
- Automated precision assembly: fill, seal, mount Kresling reservoirs in scaffold
- Device completion: cannula integration, adhesive pad bonding, hub attachment
- Terminal sterilisation (Gamma or EtO) + sterile barrier packaging
- Ensures shelf-life stability and ready-to-use clinical deployment

# Estimated Cost Per Unit (Prototyping and High Volume Manufacturing)

| Cost Component                          | Details                                              | Estimated Cost Per Unit (Low Volume / Prototype) | Estimated Cost Per Unit (High Volume / Mass Production) |
|-----------------------------------------|------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------|
| <b>A. Raw Materials</b>                 |                                                      |                                                  |                                                         |
| Silicone Kresling Units (27x)           | Medical-Grade Liquid Silicone Rubber (LSR)           | \$7.00 - \$18.00                                 | \$0.80 - \$2.50                                         |
| Polycarbonate Honeycomb                 | Medical-Grade PC                                     | \$10.00 - \$25.00                                | \$1.00 - \$3.00                                         |
| Casing                                  | Thermoplastic Polyurethane Bladder                   | \$5.00 - \$12.00                                 | \$1.50 - \$3.00                                         |
| Actuator Components                     | Ni <sub>80</sub> Fe <sub>20</sub> (Permalloy plates) | \$25.00 - \$50.00                                | \$0.50 - \$5.00                                         |
|                                         | Copper and PCB (micro-planar coils)                  | \$1.00 - \$2.00                                  | \$0.10 - \$0.50                                         |
| Drug Product (26 ml)                    | Cost of the Active Pharmaceutical Ingredient (API)   | Excluded (Highly Variable)                       | Excluded (Highly Variable)                              |
| <b>B. Manufacturing</b>                 |                                                      |                                                  |                                                         |
| Moulding / Machining                    | Machine time & labour                                | \$15.00 - \$40.00 (CNC)                          | \$0.75 - \$2.50 (Moulding)                              |
| Assembly & Filling                      | Cleanroom labour for 27 units                        | \$6.00 - \$15.00                                 | \$1.50 - \$4.00                                         |
| SLS 3D printing                         | Includes set up fees and clean up                    | \$15.00 - \$75.00                                | \$5.00 - \$12.00                                        |
| <b>C. Packaging &amp; Sterilisation</b> |                                                      |                                                  |                                                         |
| Sterile Pouch & Labelling               | Per unit cost                                        | \$1.00 - \$2.00                                  | \$0.25 - \$0.75                                         |
| Sterilisation Service                   | Per unit cost (e.g., Gamma, EtO)                     | \$0.75 - \$1.75                                  | \$0.15 - \$0.50                                         |
| Subtotal:                               | (Excluding Drug Cost)                                | ~\$85.75 - \$240.75                              | ~\$11.55 - \$33.75                                      |

# Sustainability Practices



- **Core reusable components**

- IoT Hub: For repeated use across multiple treatment cycles
- Protective Casing made of TPU to reduce device replacement

- **Modular Drug Cartridge**

- 26 mL (26 reservoirs in total) of drug can be administered as needed (at once or over longer period)
- Reduced electronic and casing waste and minimal plastic waste with reservoir design

# Device Dimensions

## Needle / Cannula System

- Gauge: **27G**
- Length: **10 mm (depth-limited insertion)**
- Soft cannula connected to hub and honeycomb cells

## Hub / Base Device (IoT Unit)

- Footprint:  **$\sim 70 \times 55$  mm**
- Maximum height above skin:  **$\sim 20$  mm**
- Mass:  **$\leq 80$  g**

## Arch / Clearance Window

- Inner window:  **$\sim 62 \times 52$  mm**
- Height:  **$\sim 16$  mm**

## Drug Cartridge / Reservoir

- Configuration:  **$3 \times 3 \times 3$  KEH honeycomb cells**
- Dimensions:  **$\sim 45 \times 55 \times 12\text{--}14$  mm**

# Overview of the Design Idea

## Core Structure

- IoT-enabled base device with replaceable honeycomb drug chamber
- Wireless charging + app connectivity

## Honeycomb Drug Chamber

- 26 kresling-embedded honeycomb cells (1 mL each)
- Each cell is connected to a microcannula
- 13 cells: hydrophilic drugs
- 13 cells: hydrophobic drugs
- 1 cell: 27G needle (10 mm) connected to main cannula
- Each chamber has a one-way valve to ensure there is no flowback

## Drug Release Mechanism

- Each kresling unit embedded within a honeycomb cell collapses individually via rotating magnetic field
- One-way valve ensures drug only releases during activation
- Cannula system:
  - Main cannula » splits into hydrophilic & hydrophobic channels
  - Each channel is connected to individual honeycomb cells

## Safety & Reusability

- Outer protective shell to prevent damage
- Replaceable drug chamber + needle » reusable IoT unit
- Light indicators:
  - Needle insertion/retraction
  - Start/stop of drug delivery
  - System issues, low battery, battery level

## Smart Features

- Body-contact sensors » send data to companion app
- Real-time updates for patients & healthcare providers

# Drug Storage and Actuation Visual Aid

3D model: [https://drive.google.com/file/d/18i13pFsIJJs4YX7hvR6EfNNSY3Yvv\\_-PM/view?usp=sharing](https://drive.google.com/file/d/18i13pFsIJJs4YX7hvR6EfNNSY3Yvv_-PM/view?usp=sharing)

